

RECO:

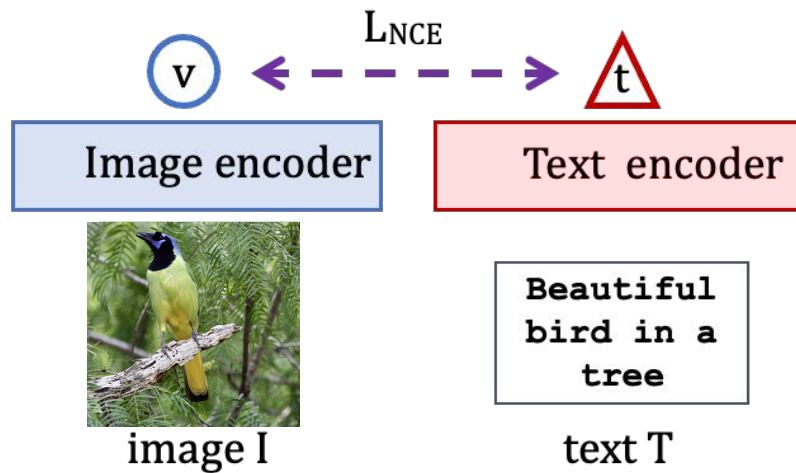
Retrieval-Enhanced Contrastive Vision-Text Models

Ahmet Iscen, Mathilde Caron, Alireza Fathi, Cordelia Schmid

ICLR 2024

Motivation

- Contrastive dual-encoder models are good describing generic objects
- They struggle on fine-grained recognition



Generic classification



What is
the object
in this
image?

Class Vocabulary

- Car
- Truck
- Bird
- Cat
- ...

Generic classification



Classify



What is
the object
in this
image?

Class Vocabulary

- Car
- Truck
- Bird
- Cat
- ...

Generic classification



Classify

Bird

What is
the object
in this
image?

Class Vocabulary

- Car
- Truck
- Bird
- Cat
- ...

Fine-grained classification



Classify

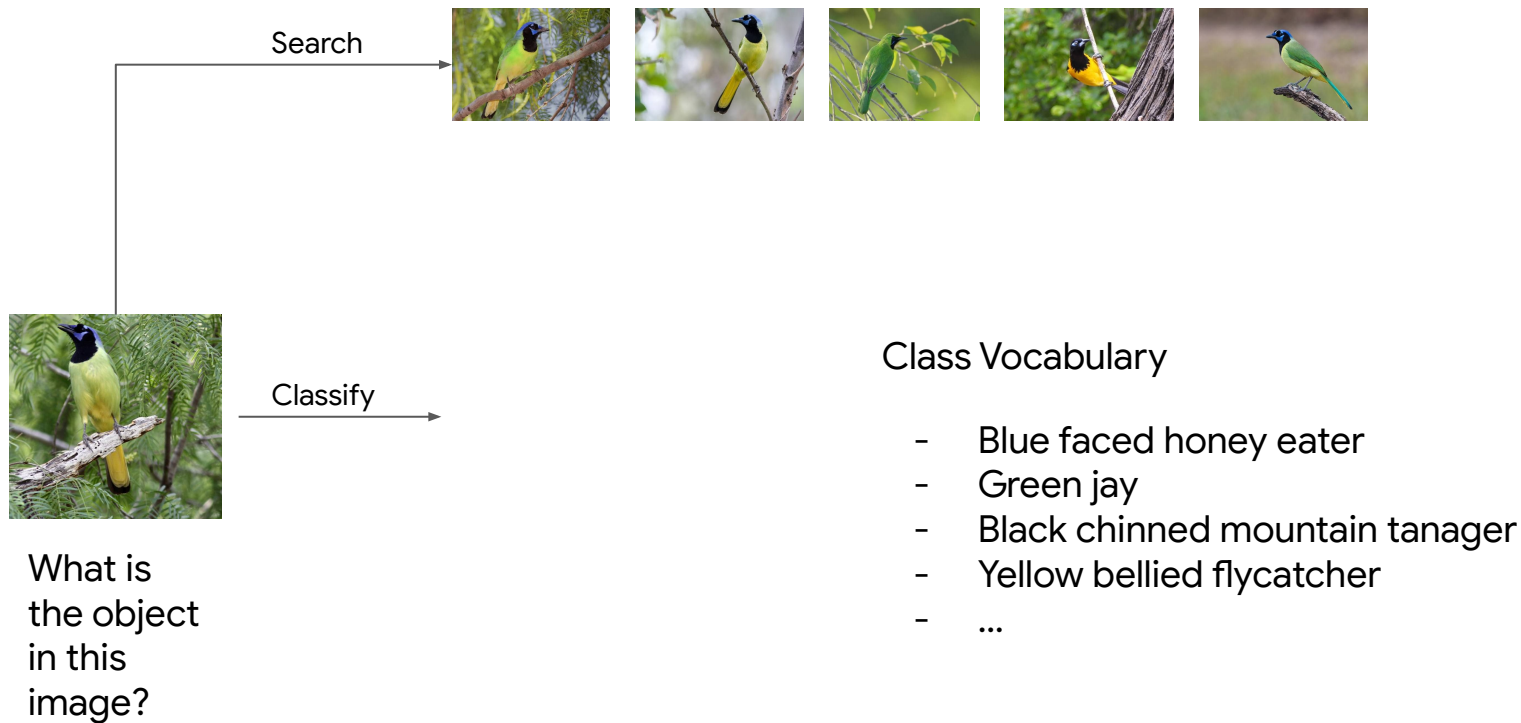


What is
the object
in this
image?

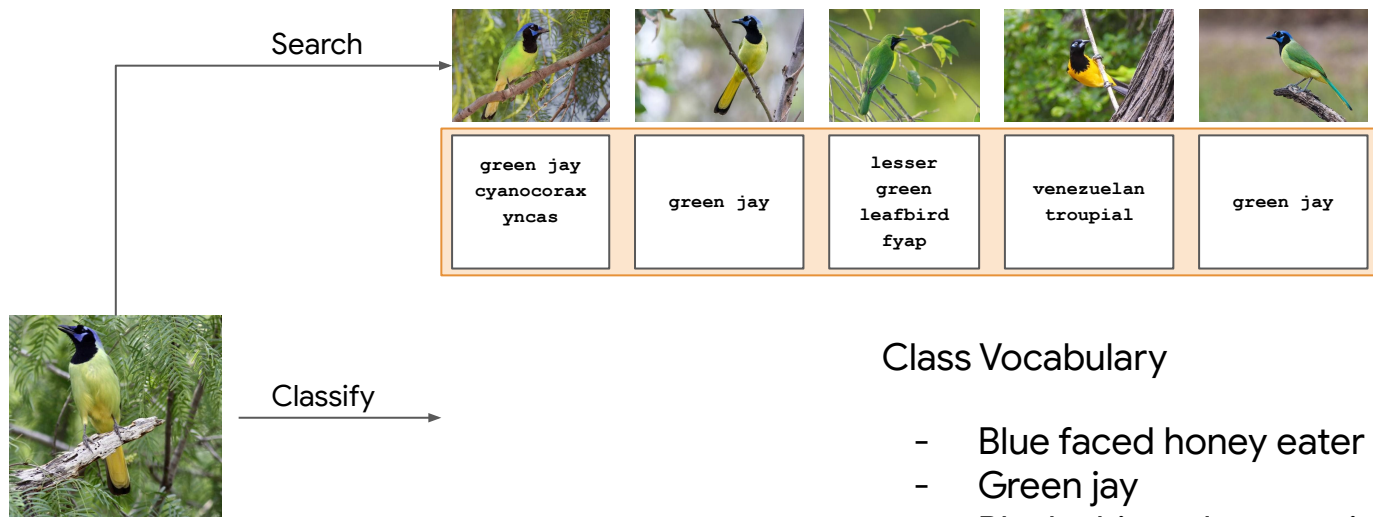
Class Vocabulary

- Blue faced honey eater
- Green jay
- Black chinned mountain tanager
- Yellow bellied flycatcher
- ...

Fine-grained classification with retrieval



Fine-grained classification with retrieval

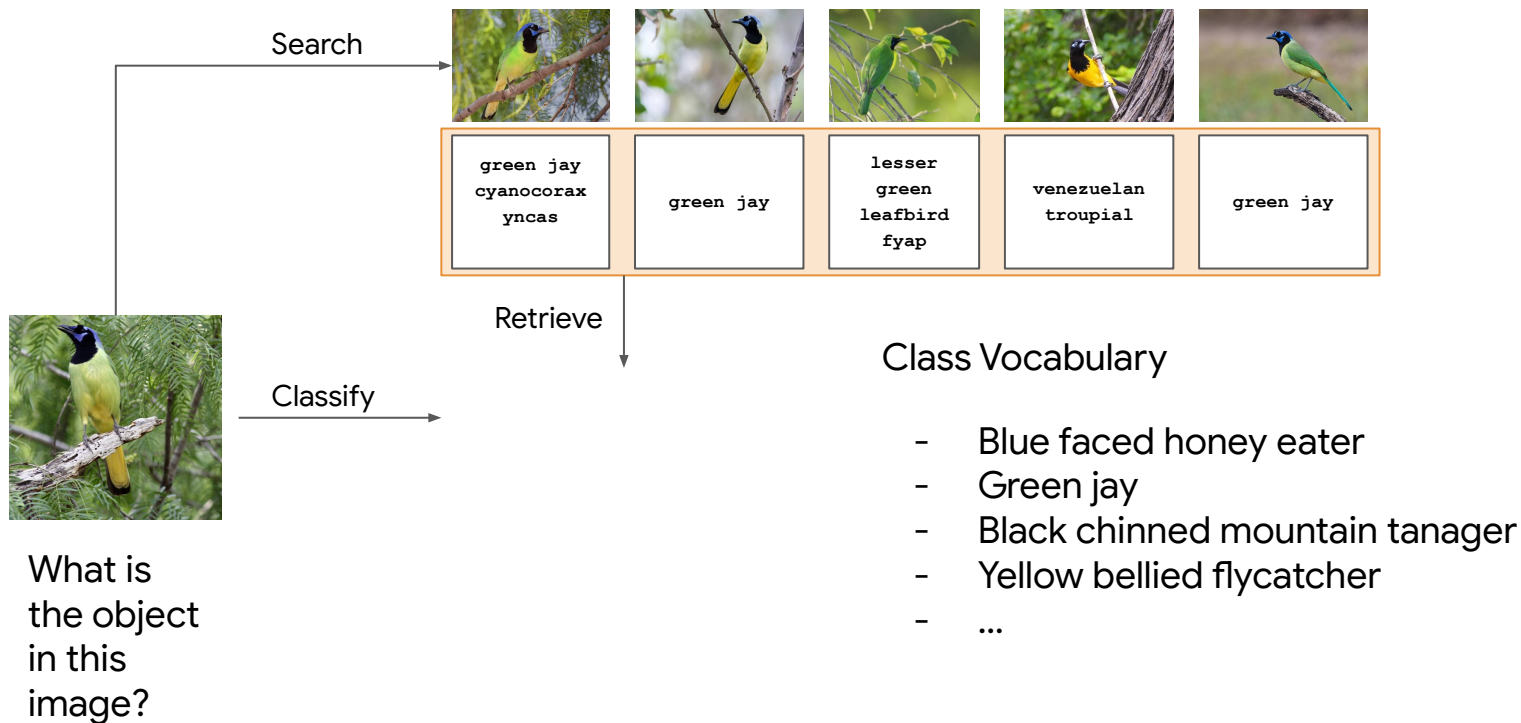


What is
the object
in this
image?

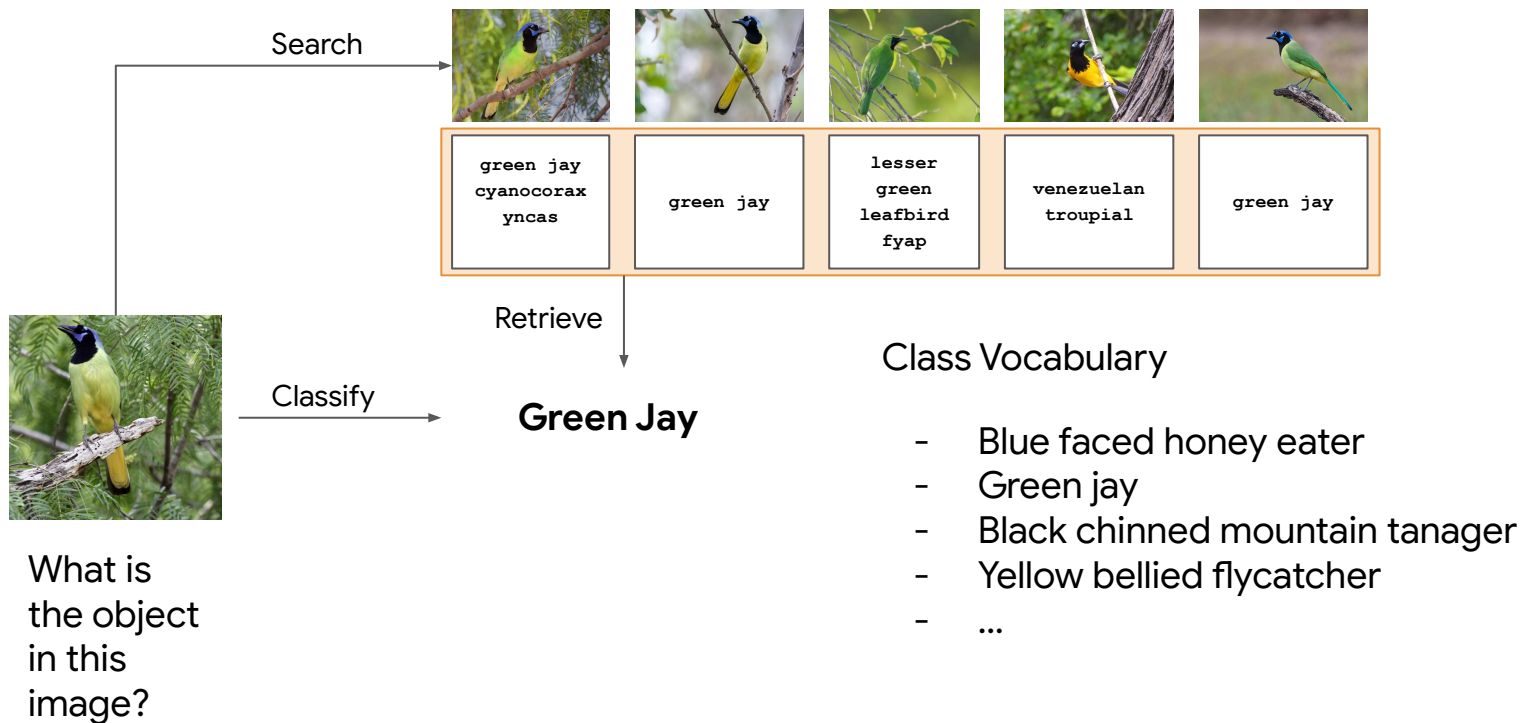
Class Vocabulary

- Blue faced honey eater
- Green jay
- Black chinned mountain tanager
- Yellow bellied flycatcher
- ...

Fine-grained classification with retrieval



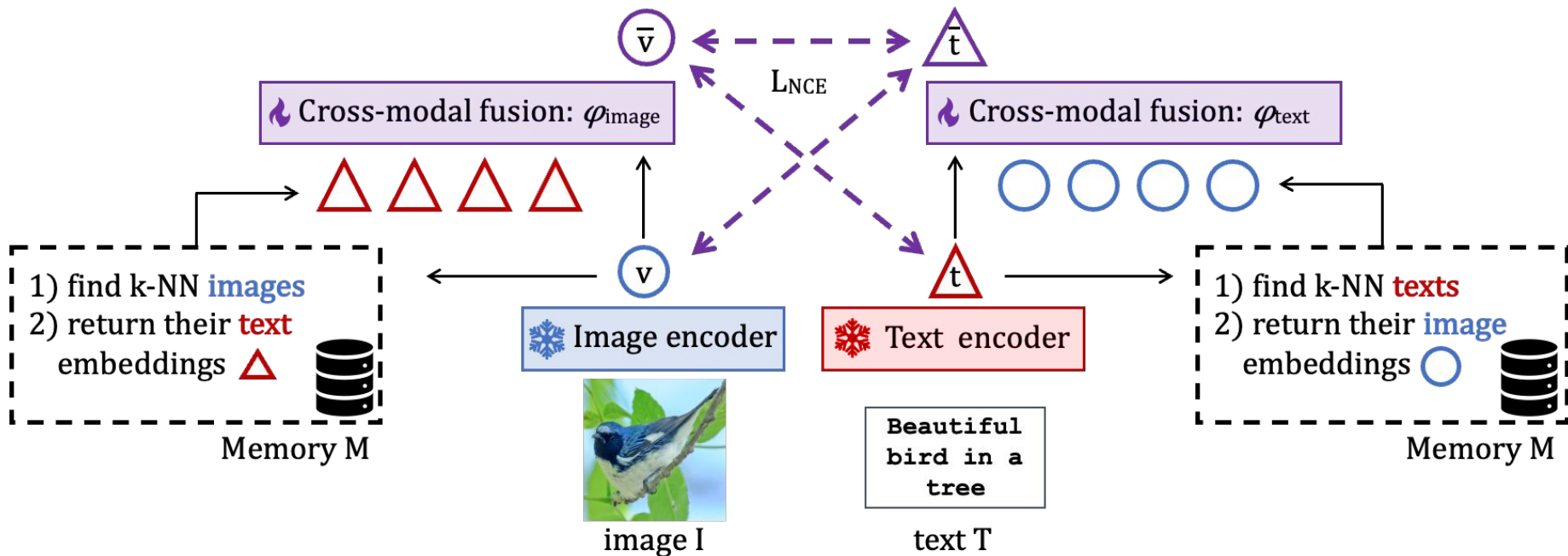
Fine-grained classification with retrieval



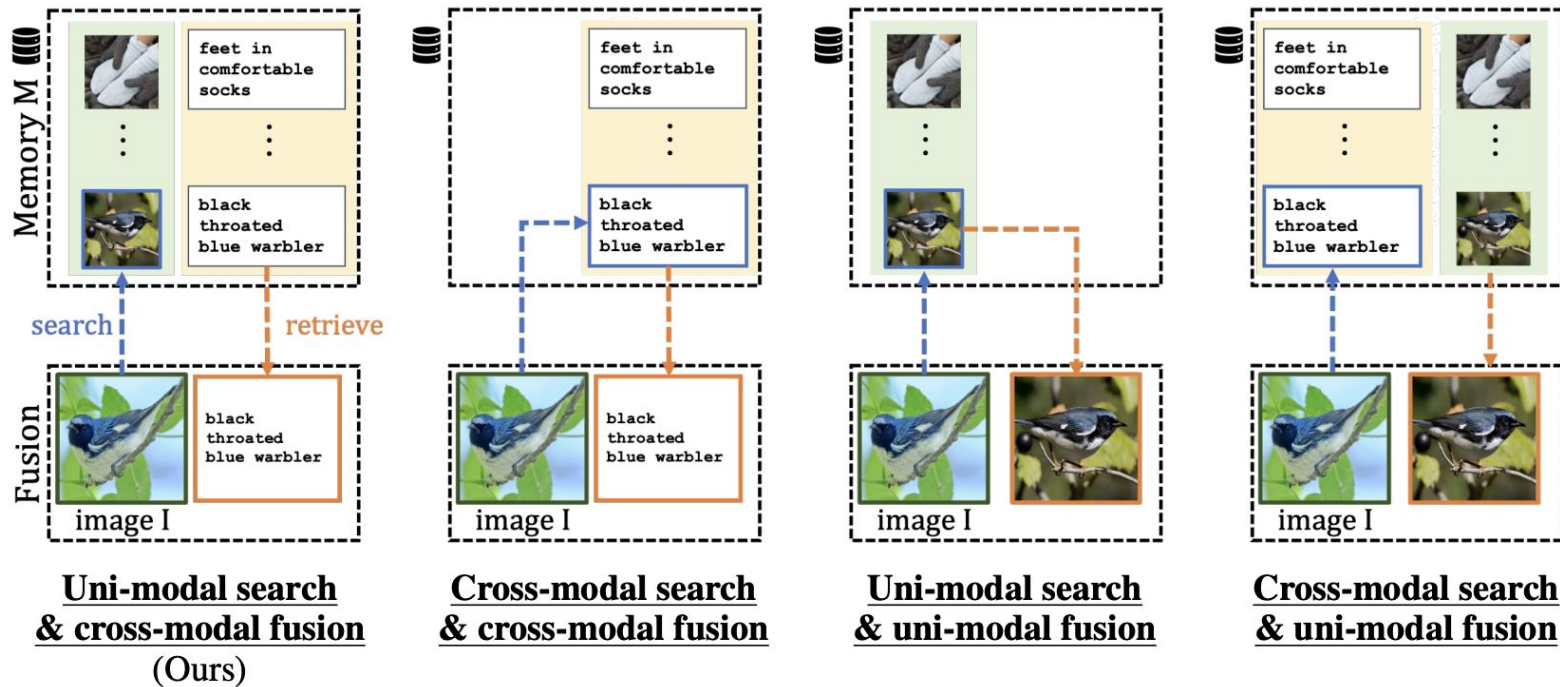
Method

- Start with a pre-trained dual-encoder model
- Freeze the encoders and learn a retrieval module
- Use the retrieved data at inference

Overview



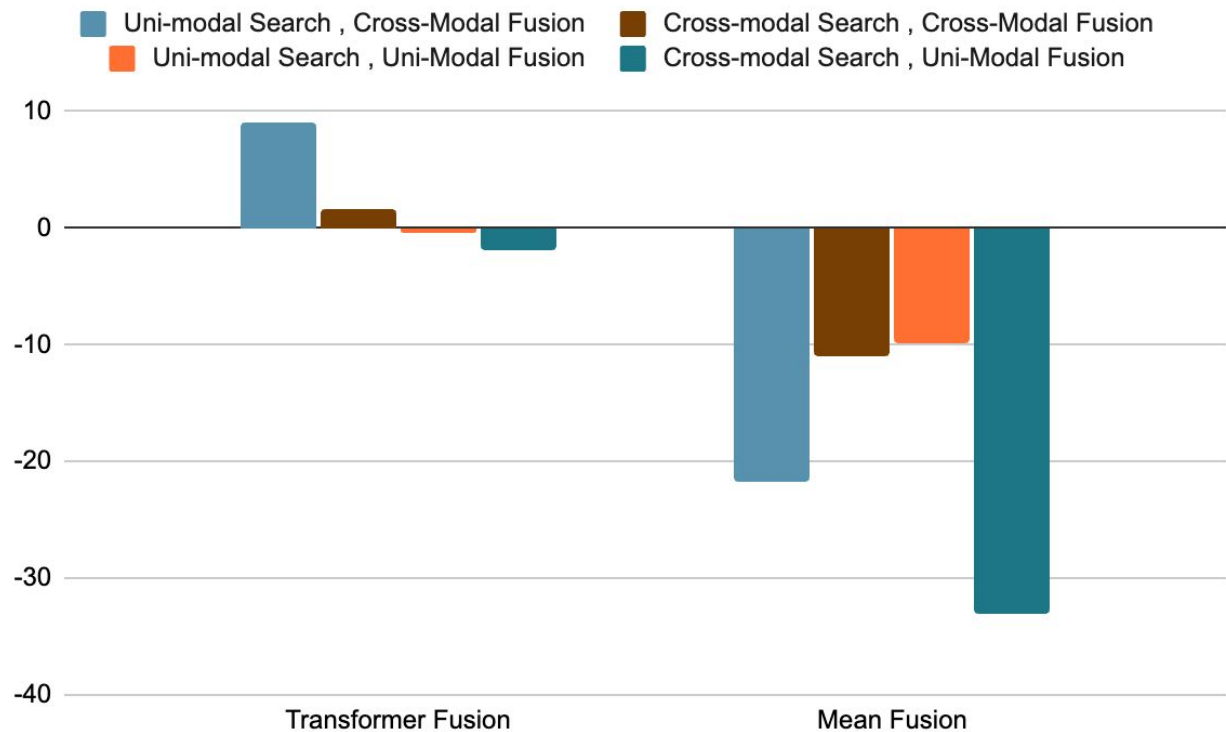
Search and Fusion



Experiments

- Start from frozen CLIP
- Train the retrieval module on CC12M (11M image-text pairs)
 - Memory: Webli, 1B image-text pairs
- Evaluate the trained model in zero-shot downstream tasks
 - Stanford Cars, CUB (Birds), Oxford Flowers, ImageNet-1k, Places365, Flickr30k (retrieval), COCO Captions (retrieval)
- Zero-shot evaluation
 - Embed class names
 - Embed query images
 - Do similarity search

Relative Difference wrt No Retrieval



Qualitative Examples

Query	<u>Uni</u> -modal search (Ours)	<u>Cross</u> -modal search
	retrieved captions <u>green jay</u> cyanocorax yncas <u>green jay</u> lesser green leafbird fyap search v → v   	retrieved captions a front view of a black chinned mountain tanager a blue faced honeyeater in adelaide zoo a yellow legged honeycreeper search v → t   
	retrieved captions 2005 <u>chrysler crossfire</u> for sale at auto quest usa inc 2005 <u>chrysler crossfire</u> 2006 <u>chrysler crossfire</u> for sale in fort myers beach fl search v → v   	retrieved captions 2003 mercedes benz s155 convertible image 2003 mercedes benz s1500 convertible image 2003 mercedes benz s1500 convertible search v → t   

Qualitative Examples

Query	<u>Uni</u> -modal search (Ours)	<u>Cross</u> -modal search
Yellow bellied Flycatcher	retrieved images  search t → v yellow bellied flycatcher photo yellow bellied flycatcher yellow bellied flycatcher	retrieved images  search t → v tropical kingbird grey headed canary flycatcher yellow breasted chat
Daewoo Nubira Wagon 2002	retrieved images  search t → v daewoo nubira wagon ii 2003 daewoo nubira wagon ii 2001 daewoo nubira wagon ii 2004 05	retrieved images  search t → v speedage hyundai santro dlx p back do 30 1999 daewoo nubira 1 1996 toyota granvia kch16w 264 uss sapporo

Experiments

Method	# par.	Image classification						T→I		I→T	
		Cars	CUB	Flowers	Im1k	Pl365	Dogs	COCO	Flickr	COCO	Flickr
CLIP-B/32	151	57.2	52.8	62.1	63.5	40.6	58.6	30.2	61.1	51.2	80.9
+ RECO	154	68.1(+10.9)	63.0(+10.2)	67.9(+5.8)	64.6(+1.1)	42.2(+1.6)	59.7(+1.1)	33.6(+3.4)	65.7(+4.6)	52.2(+1.1)	81.8(+0.9)
CLIP-L/14	428	75.6	61.7	75.6	75.5	42.0	72.7	35.2	68.6	57.2	87.5
+ RECO	435	82.8 (+7.2)	73.4 (+11.7)	79.5 (+3.9)	76.1 (+0.6)	43.6(+1.6)	73.9 (+1.2)	38.7(+3.5)	72.6 (+4.0)	58.0 (+0.8)	88.5 (+1.0)
<i>Other approaches</i>											
K-Lite [44]	151	10.0	—	78.6	52.3	—	—	—	—	—	—
Flava [45]	172	—	—	—	—	—	—	38.4	65.2	42.7	67.7
Align [23]	247	78.7	38.2	64.9	67.6	44.0	56.3	40.2	72.6	55.1	86.7
LiT-L [52]	638	24.8	61.7	74.1	75.8	42.6	70.8	31.1	57.6	48.5	77.7
PaLI [7]	17,000	—	—	—	72.1	—	—	—	—	—	—

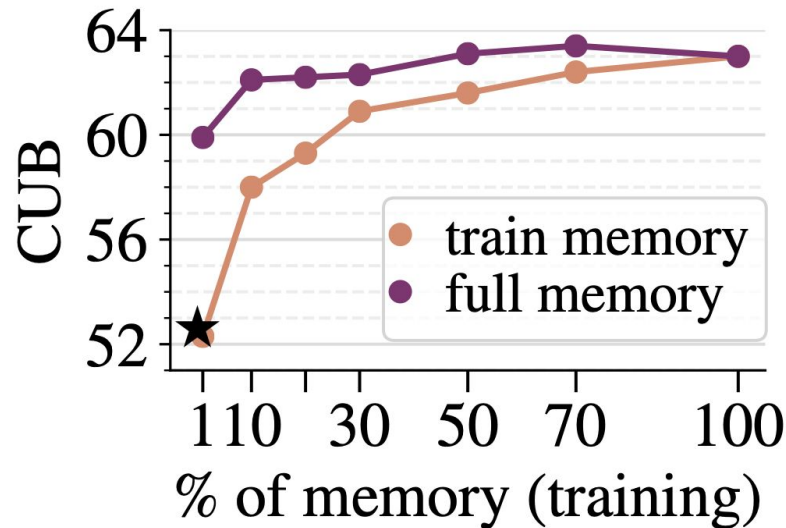
Experiments - OVEN Benchmark

- More than 6M entities collected from Wikipedia
- No fine-tuning
 - Zero-shot

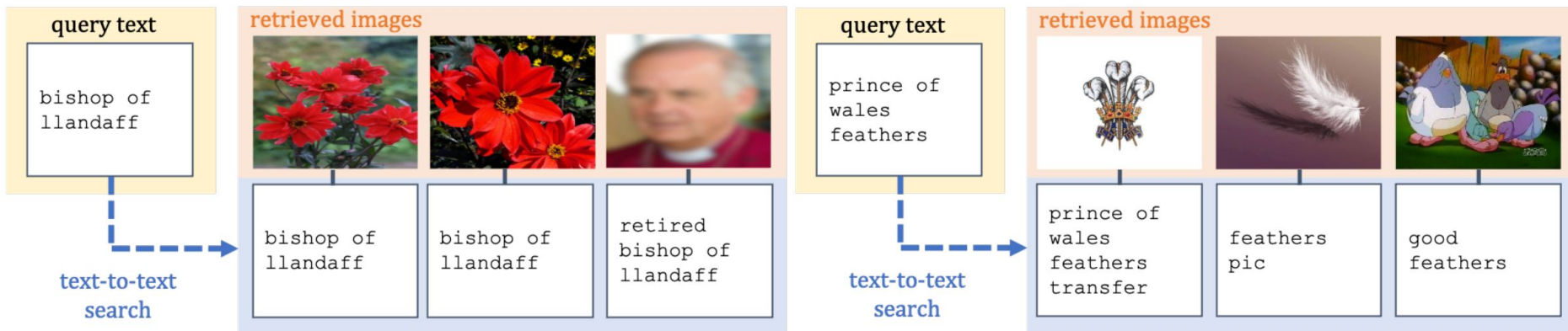
Method	# params (M)	Seen	Unseen	Harmonic mean
<i>Zero-shot</i>				
PaLI-17B [7]	17,000	4.4	1.2	1.9
CLIP-L/14 [41]	428	5.6	4.9	5.3
CLIP-L/14 [41] + RECO (Ours)	435	12.4 (+6.8)	12.7 (+7.8)	12.6 (+7.3)
<i>Fine-tuning on the OVEN Seen categories</i>				
CLIP-L/14 Fusion [19]	880	33.6	4.8	8.4
PaLI-3B [7]	3,000	19.1	6.0	9.3
CLIP-L/14 CLIP2CLIP [19]	860	12.6	10.5	11.5
PaLI-17B [7]	17,000	28.3	11.2	16.1

Impact of the Memory

- Train on smaller memory
- Use the full memory on inference



Failure cases



Conclusions

- Retrieving external data helps improve zero-shot fine-grained classification
- Search: Uni-modal similarity is important
- Fusion: Cross-modal fusion is important

GERALD:

A Generative Approach for Wikipedia-Scale Visual Entity Recognition

Mathilde Caron, Ahmet Iscen, Alireza Fathi, Cordelia Schmid

CVPR 2024

Overview

- Problem definition
- Traditional approaches and related work
- Our method GER-ALD
- Entity-based pretraining
- Results
- Conclusion

Problem: OVEN webscale visual entity recognition

Input = (image + text)



What is the main object?

→
visual entity
recognition

Entity: Q358813
Name: Black-and-white colobus

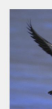


Entity vocabulary:
>6M entities



WIKIPEDIA
The Free Encyclopedia

Entity: Q845292
Name: American black duck



Entity: Q521977
Name: Black c



Entity: Q358813
Name: Black-and-white colobus



Entity: Q2638325
Name: Puente de Piedra (Zaragoza)



Entity: Q3983228
Name: Rolls-Royce Phantom (seventh generation)



Entity: Q18164401
Name: Saint John the Baptist Parish Church (Liliw)



Traditional approaches for visual entity recognition

- Classification
 - Problem: does not scale
- Dual encoder (CLIP or ALIGN)
 - Problem: struggles for fine-grained recognition
 - Problem: discrepancy between training and inference
- Captioning models (PaLI or GIT)
 - Problem: discrepancy between captions and entities

Related works in NLP

To overcome limitations of dual encoder approaches, recent work in NLP explores generative retrieval.

- [Transformer Memory as a Differentiable Search Index](#)
- [Recommender Systems with Generative Retrieval](#)
- [A Neural Corpus Indexer for Document Retrieval](#)
- [How Does Generative Retrieval Scale to Millions of Passages?](#)

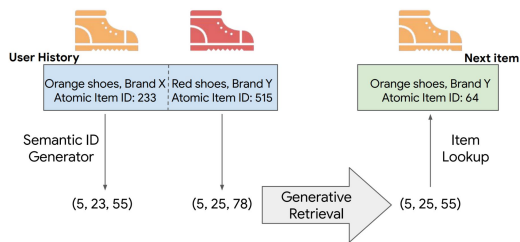


Figure 1: Overview of the *Transformer Index for Generative Recommenders* (TIGER) framework. With TIGER, sequential recommendation is expressed as a generative retrieval task by representing each item as a tuple of discrete semantic tokens.

Overview



Problem definition



Traditional approaches and related work



Our method GER-ALD



Generative Entity Recognition (**GER**)

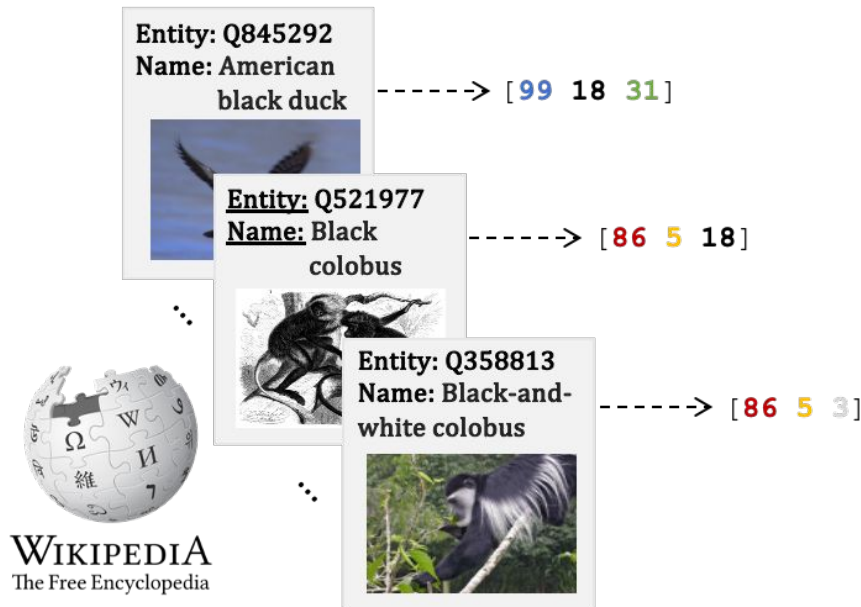


Creating unambiguous language-based discriminative codes (**ALD**)

- Entity-based pretraining
- Results
- Conclusion

New paradigm: generative entity recognition (GER)

Creating semantic compact codes for 6M Wikipedia Entities



Code creation

We want the codes to be:

- **Semantically-structured**
 - Similar entities should have common code tokens
- **Discriminative and compact**
 - The code should represent the entity efficiently without extra non informative tokens
- **Unambiguous**
 - Each entity has an unique code

Code creation: Language-based

Entity: Q358813
Name: Black-and-white colobus



Step 1: entity name
text tokenization



[black] [-] [and] [-] [white] [col] [ob] [us]

Entity: Q521977
Name: Black colobus



[black] [col] [ob] [us]

Code creation: Language-based

Entity: Q358813
Name: Black-and-white colobus



Step 1: entity name
text tokenization



[black] [-] [and] [-] [white] [col] [ob] [us]

Entity: Q521977
Name: Black colobus



[black] [col] [ob] [us]

By using language-based codes, similar entities naturally share subsets of code tokens.

Code creation: Language-based

Entity: Q358813
Name: Black-and-white colobus



Step 1: entity name
text tokenization



[18] [19] [2] [19] [3] [86] [5] [100]

Entity: Q521977
Name: Black colobus



[18] [86] [5] [100]

By using language-based codes, similar entities naturally share subsets of code tokens.

Code creation: Discriminative

Not all information is needed in the tokenized entity name.

- The species “colobus” could be represented by 2 tokens ([col] [ob]) instead of 3 ([col] [ob] [us])
- Some tokens are not informative: [-] [and] [-]

Code creation: Discriminative

Not all information is needed in the tokenized entity name.

- The species “colobus” could be represented by 2 tokens ([col] [ob]) instead of 3 ([col] [ob] [us])
- Some tokens are not informative: [-] [and] [-]



[black] [-] [and] [-] [white] [col] [ob] [us]

Step 2: rank tokens
by frequency

[col] [ob] [white] [black] [us] [-] [-] [and]



[black] [col] [ob] [us]

Step 2: rank tokens
by frequency

[col] [ob] [black] [us]

Code creation: Discriminative

Entity: Q358813
Name: Black-and-white colobus



[black] [-] [and] [-] [white] [col] [ob] [us]

Step 3: keep the
rarest tokens only

[col] [ob] [white] [b~~l~~ack] [~~us~~] [~~and~~] [~~-~~] [~~-~~]

We remove the
clutter information

Entity: Q521977
Name: Black colobus



[black] [col] [ob] [us]

Step 3: keep the
rarest tokens only

[col] [ob] [black] [~~us~~]

Code creation: Discriminative

Entity: Q358813
Name: Black-and-white colobus



[black] [-] [and] [-] [white] [col] [ob] [us]

Step 3: keep the
rarest tokens only



[86] [5] [3]

Entity: Q521977
Name: Black colobus



[black] [col] [ob] [us]

Step 3: keep the
rarest tokens only



[86] [5] [18]

Code creation: unAmbiguous

When several entities have the same code, we use an extra token to differentiate them.

Entity: Q244947
Name: Aqueduct of Segovia



[duct] [que] [a] [seg~~x~~ia] [~~x~~f]

Entity: Q2842746
Name: Aqueduct

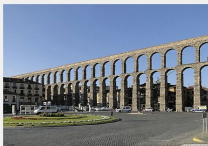


[duct] [que] [a]

Code creation: unAmbiguous

When several entities have the same code, we use an extra token to differentiate them.

Entity: Q244947
Name: Aqueduct of Segovia



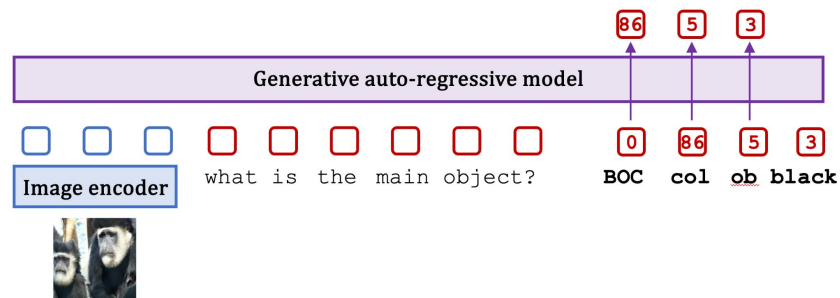
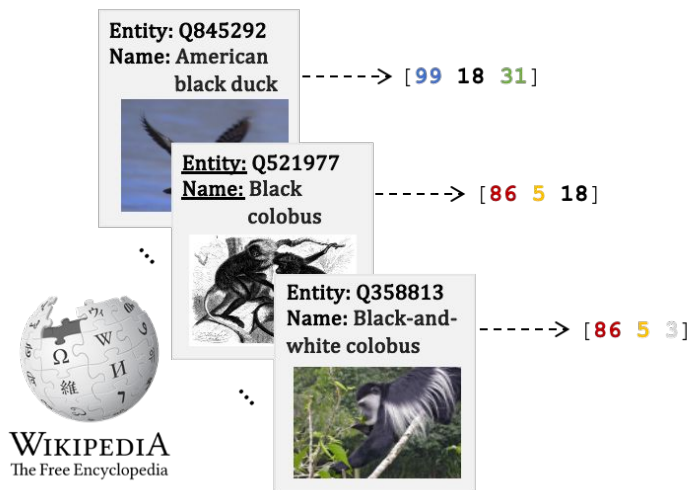
[duct] [que] [a] [segovia] [X]

Entity: Q2842746
Name: Aqueduct



[duct] [que] [a]

GER-ALD models use unAmbiguous Language-based Discriminative codes



Overview

- ✓ Problem definition
- ✓ Traditional approaches and related work
- ✓ Our method GER-ALD
- 🔄 Entity-based pretraining
 - Results
 - Conclusion

Entity-based pre-training

- Traditional captioning pre-training does not transfer well to entity-recognition
- Zero-shot predictions of GiT-Large* pre-trained on WebLI:



What is this plant?

Target:

"Euphorbia marginata."

Zero-shot prediction:

"I'm not sure if it's a flower."



Which category of object is shown in the image?

Target:

"Chain mail."

Zero-shot prediction:

"Leg."



What is the main object ?

Target:

"Blue crab."

Zero-shot prediction:

"Blue crab ? You need to know."

*: GiT-Large is similar to PaLI but with less parameters (400M)

Entity-based pre-training

- Finetuning on OVEN does help but we would like to improve on the pretraining front directly



What is this plant?

Target:
"Euphorbia marginata."

Zero-shot prediction:
"I'm not sure if it's a flower."

Finetuned prediction:
"Clematis."



Which category of object is shown in the image?

Target:
"Chain mail."

Zero-shot prediction:
"Leg."

Finetuned prediction:
"Chain mail."



What is the main object ?

Target:
"Blue crab."

Zero-shot prediction:
"Blue crab ? You need to know."

Finetuned prediction:
"Blue crab."

Entity-based pre-training

- We use entity-name to caption similarity to retrieve relevant images for each of the OVEN entities.

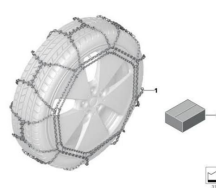
Q747361

Wikipedia title: Snow chains

Wikipedia img:



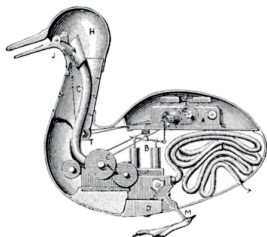
Imgs retrieved
from WebLi



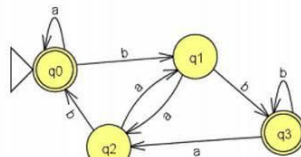
Q787116

Wikipedia title: Automaton

Wikipedia img:



Imgs retrieved
from WebLi



Experimental setup

- OVEN dataset has 6,063,945 different entities
 - Training set contains only a subset of the entities: the “seen” entities.
 - We report harmonic mean (HM) between seen and unseen entities.
-
- Step1) Pretraining on WebLI
 - Step2) Finetuning on OVEN train set (only seen entities)

Entity-based pre-training

We report performance after pretraining (no finetuning)

Pretraining dataset selection from WebLI	Codes	HM
Random	Original caption	1.8

Very poor performance for zero-shot models pretrained with captioning on WebLI



Entity-based pre-training

Pretraining dataset selection from WebLI	Codes	HM
Random	Original caption	1.8
Caption sim. w/ OVEN entities	Original caption	12.9 (+11.1)

Subset of WebLI is more relevant to the OVEN entities !

Entity-based pre-training

Pretraining dataset selection from WebLI	Codes	HM
Random	Original caption	1.8
Caption sim. w/ OVEN entities	Original caption	12.9 (+11.1)
Caption sim. w/ OVEN entities	Entity name	14.8 (+1.9)

Predicting the entity name
rather than the original WebLI
captions works better



Original caption:

Discover muskellunge facts about the state fish of Wisconsin.

Entity name:

Muskellunge



Original caption:

School zones have a speed limit of 20 mph which is in effect during school days from 7 in the morning till 5 in the evening [...]

Entity name:

School zone

ALD codes versus tokenized entity names

Pretraining dataset selection from WebLI	Codes	HM
Random	Original caption	1.8
Caption sim. w/ OVEN entities	Original caption	12.9 (+11.1)
Caption sim. w/ OVEN entities	Entity name	14.8 (+1.9)
Caption sim. w/ OVEN entities	ALD	17.5 (+2.7)

ALD codes are better than predicting the tokenized entity names (especially true for entities with long names...)

Overview

- ✓ Problem definition
- ✓ Traditional approaches and related work
- ✓ Our method GER-ALD
 - ✓ Generative Entity Recognition (**GER**)
 - ✓ Creating unambiguous language-based discriminative codes (**ALD**)
- ✓ Entity-based pretraining
- 🔄 Results
 - 🔄 Comparison with SOTA
 - 🔄 Comparison with alternative code creation strategies
- Conclusion

Comparison with state-of-the-art

Method	#par.(B)	Pretraining		OVEN test		
		dataset	#imgs	HM	seen	unseen
<i>Dual encoder approaches</i>						
CLIP _{ViT-L14}	0.42	OpenAI	400M	5.2	5.6	4.9
CLIPfusion _{ViT-L14}	0.88	OpenAI	400M	8.4	33.6	4.8
CLIP2CLIP _{ViT-L14}	0.86	OpenAI	400M	11.5	12.6	10.5
<i>Captioning approaches</i>						
GiT-Large	0.40	WebLI	100M	7.0	17.6	4.3
PaLI-3B	3	WebLI	1B	9.1	19.1	6.0
PaLI-17B	17	WebLI	1B	16.0	28.3	11.2
<i>Generative entity recognition</i>						
GER-ALD [‡] (Ours)	0.40	Entity-LAION	41M	20.9	29.1	16.3
GER-ALD (Ours)	0.40	Entity-WebLI	55M	22.7	31.5	17.7

GER-ALD improves over PaLI-17B while using 42x less parameters.

Comparison with other code creation strategies

Method	Entity-based pretraining			+ finetuning on seen		
	HM	Seen	Unseen	HM	Seen	Unseen
GER-ATOMIC	15.9	15.3	16.7	20.1	26.2	16.3
GER-CAPTION	14.3	16.5	12.6	20.7	26.8	16.9
GER-HKC	15.8	15.5	16.0	21.0	25.2	17.9
GER-ALD	17.6	18.0	17.2	22.8	32.7	17.5

Conclusion

- Novel generative framework for web-scale visual entity recognition
- We represent each entity by a compact, discriminative and semantic code that a generative image-to-text model learns to auto-regressively decode.
- Future directions:
 - better entity codes by leveraging additional information?
 - Wikipedia page content and/or its attached image
 - by using external tools
 - Focus on the query split of OVEN? Importance of the question input ++
 - Uncertainty ?

Thank you for your attention!

